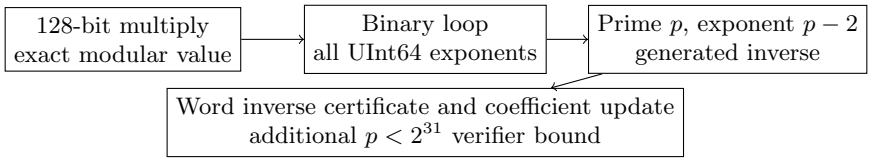
Widened modular powers and generated inverse certificates 2026-10-31

T-1154. Producer source-model proof; independent source-fidelity review pending.

Checked chain

For arbitrary UInt64 operands, $ab < 2^{128}$. The BitVec128 model of widening, multiplying, reducing and narrowing equals exact natural modular multiplication for positive modulus. Its remainder fits back into UInt64; this primitive needs no $p < 2^{31}$ bound.

The binary-loop invariant is $r\,a^e$. Parity, right shift and skipped final squaring preserve it. The fuel theorem assumes $e < 2^{\text{fuel}}$, so 64 steps cover every UInt64 exponent. For $1 < p$ and canonical base, the result is canonical and decodes to a^e.



For prime p and nonzero canonical input, the standard finite-group-cardinality argument gives the inverse. Under $p < 2^{31}$, its word product supplies the preceding InverseWitness. The composed coefficient-update theorem now uses this generated inverse.

Controls and readback

Fourteen loop declarations, six inversion/composition declarations and eighteen controls receive 38 named axiom readbacks. Controls include inverse $2 \mapsto 9$ modulo 17, identity/negative-one inverses, exponents $0/1/2^{64}-1$, zero base, maximum operands and the composed slot result.

The widened value $2^{63} \cdot 2 \bmod 5$ is 1; narrow UInt64 multiplication gives 0. At composite $p = 15, a = 2$, exponent $p-2$ returns 2 and fails the inverse certificate. Zero modulus, modulus one and zero inversion input are excluded from their respective theorem domains.

Source and evidence

Source excerpts/hashes bind `three_leads.rs`:195–200,307–332,588–600. The source dispatch is odd-characteristic $n = 1$ without diagnostic tables. BitVec128 models Rust u128 arithmetic; this is not compiled-binary equivalence. Full array-loop composition and geometric class transport remain separate bindings.

The first widened attempt needed the explicit ofNat conversion lemma; the first Fermat attempt needed an implicit argument instantiated. Corrected attempts and controls passed. Nonfatal deprecation warnings remain. Two exploratory cache planners failed import parsing; no cache was copied or built. Existing imported group-cardinality lemmas sufficed. Error summaries and native Lean receipts are retained separately.

Native exits, unchanged input hashes and standard-axiom readback accompany the inspected PDF, editable diagram, status JSON and custody. Existing compiled dependencies remain in the trust boundary.

Sources: Lean Bitvectors and Mathlib finite-field/Fermat identities, compared with pinned source. No scientific campaign, enumeration, timing or cost ratio. Cohort and campaign limits unchanged.